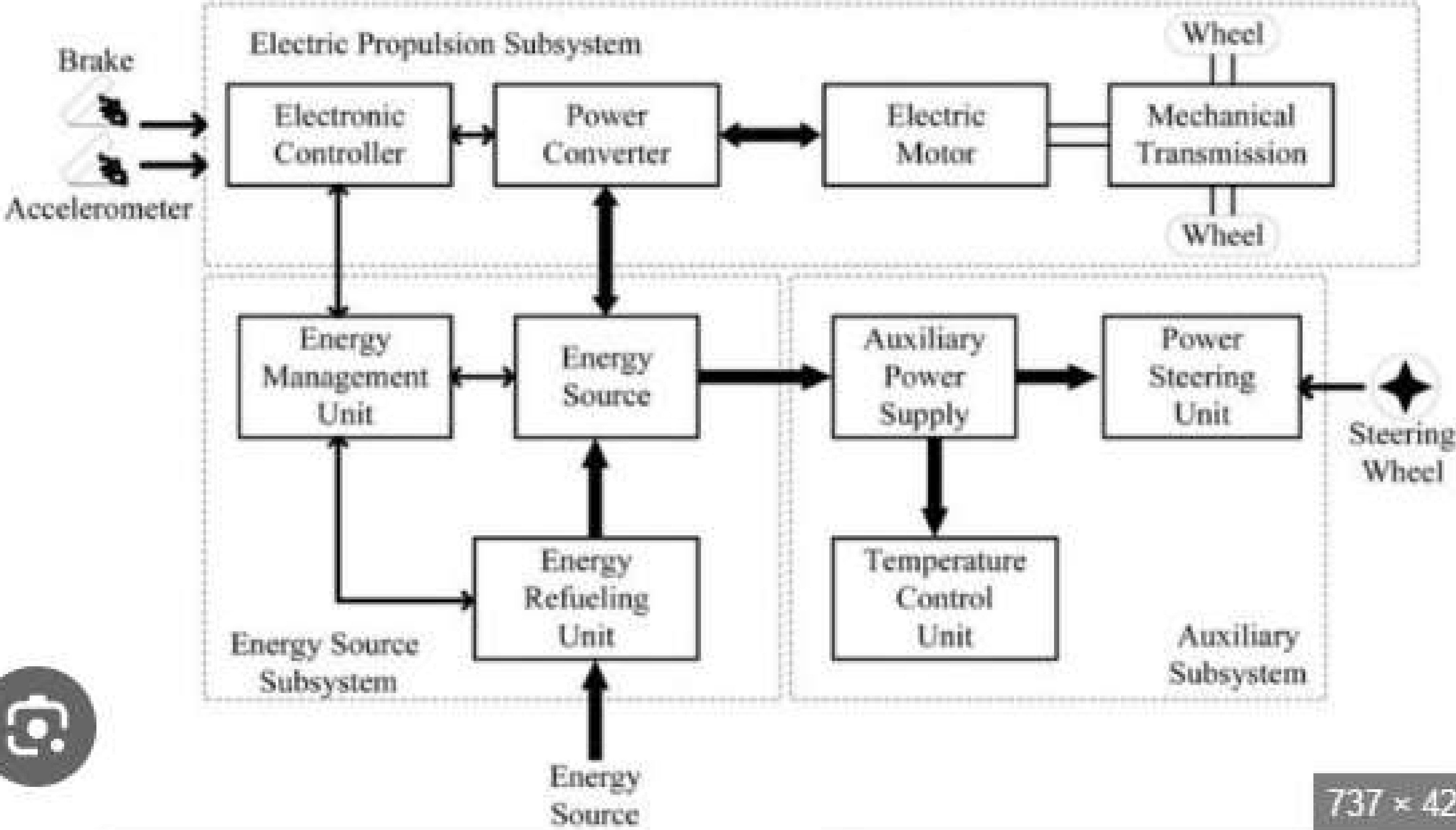
Electric Propulsion Subsystem
Brake
Accelerometer
Electronic Controller
Power Converter
Electric Motor
Mechanical Transmission
Wheel
Wheel
Energy Management Unit
Energy Source
Energy Refueling Unit
Energy Source Subsystem
Energy Source
Auxiliary Power Supply
Power Steering Unit
Steering Wheel
Temperature Control Unit
Auxiliary Subsystem

Automobile Systems & Electric Vehicles: A Compact Overview

1. Automobile Systems (ICE Vehicle)

An automobile integrates multiple mechanical and electrical systems:

- Engine – Converts fuel into motion via combustion.
- Fuel System – Stores & supplies fuel (tank, pump, injectors).
- Exhaust System – Removes gases & reduces noise.
- Cooling & Lubrication – Prevents overheating & friction.
- Transmission – Transfers power to wheels (gearbox, clutch, differential).
- Braking & Steering – Controls speed & direction.
- Electrical System – Battery, alternator, starter, lights.

These systems work together but require frequent maintenance (oil, filters, belts).

---

2. EV Components & Working

EVs replace the engine and fuel with an electric motor & battery.

Main Components:

- Traction Battery (Lithium-ion) – Stores energy.
- Electric Motor – Drives wheels.
- Inverter – Converts DC to AC.
- Onboard Charger – Charges battery from grid.
- Regenerative Braking – Recovers energy while braking.

Working:

1. Battery → Inverter → Motor → Wheels.
2. During braking → Motor acts as generator → Battery recharges.

No fuel, no exhaust, no clutch, no gearbox (in most EVs).

---

3. ICE vs EV – Quick Comparison

Parameter	ICE Vehicle	Electric Vehicle
Energy	Petrol/Diesel	Electricity
Efficiency	~20–30%	~70–85%
Fuel/Charge time	2–5 min	30 min – 8 hrs
Maintenance	High	Low
Tailpipe emission	Yes	No
Range	400–600 km	200–500 km
Upfront cost	Lower	Higher
Running cost	Higher	Lower

---

4. Advantages & Challenges of EVs

Advantages

- Zero tailpipe pollution
- Low running & maintenance cost
- Silent & smooth drive
- Instant torque (fast acceleration)
- Home charging possible

Challenges

- High initial purchase price
- Limited charging stations
- Longer “refueling” time
- Battery degrades over time
- Disposal/recycling concerns

Summary Table – Advantages & Challenges at a Glance

Category	Details
Advantages	Eco-friendly, low cost/km, less maintenance, quiet, instant torque
Challenges	High upfront cost, range anxiety, slow charging, battery life, infrastructure